KARTHIK VEERANALA

Hyderabad, India | Email
Portfolio | GitHub | LinkedIn

B.Tech CSE – IARE
CGPA: 7.5 / 10
Expected May 2027

Profile

Computer Science and Engineering undergraduate with experience in game programming, Unreal Engine development, automated testing, and interactive prototyping. Developed gameplay systems, physics-based mechanics, puzzles, and testing tools across projects and internships. Combines software engineering and problem-solving with game systems, player experience, and collaborative development.

Technical Skills

Programming: C++, C#, Python, JavaScript, TypeScript, HLSL
Game Development: Unreal Engine 5.7, Unreal Engine 4, Blueprints, Phaser 3, Unity, Slate, UMG
Gameplay: Combat systems, player movement, physics, enemy AI, puzzles, progression, gameplay feedback
Software and Testing: Test automation, UI testing, debugging, Windows API, multithreading, CI workflows
Tools: Git, GitHub, CMake, Visual Studio, RenderDoc, GitHub Actions, FFmpeg
Design: Gameplay loops, player objectives, encounter design, level mapping, prototyping, playtesting, iteration

Professional Experience

Unreal Engine Game Programmer Intern **Mar 2026 – Sep 2026**
Cyrus 365 | *Hyderabad / Remote*

- Developed automated testing workflows in Unreal Engine 5.7 to validate navigation, UI, authentication flows, and gameplay.
- Contributed to UI automation and verification tools for identifying issues in player-facing features.
- Worked on test recording and continuous-integration workflows to support repeatable verification of game systems.
- Contributed to discussions on level structure, rewards, and gameplay interactions alongside programming tasks.

Game Developer **Jun 2025 – Nov 2025**
Aicade | *Remote*

- Developed and iterated on interactive prototypes, refining mechanics, responsiveness, player feedback, and stability.
- Contributed gameplay systems to *City of Aethel*, selected for the IGDC 2025 showcase.
- Built browser-based prototypes using Phaser 3, WebGL, JavaScript, and TypeScript.
- Worked on combo input handling, dodge timing, and multi-phase boss encounters.

Projects

Unreal Engine 5.7 E2E Test Automation Framework **Cyrus 365**
C++ | *Unreal Engine 5.7* | *Test Automation*

- Developed a project-agnostic testing framework for validating Unreal Engine workflows, UI interactions, and gameplay behavior.
- Worked on automated test execution, runtime UI inspection, and recorded test evidence.

The Interlude – Zero-Gravity Space Combat **1st Place, CodeDay 2.0**
Unreal Engine 4 | *C++* | *Blueprints*

- Developed six-degree-of-freedom flight mechanics and rotational controls for zero-gravity movement.
- Implemented predictive targeting and combat interactions with visual and audio feedback.

City of Aethel – Action-Combat Game **IGDC 2025 Showcase**
Gameplay Programming | *Combat Systems* | *Level Design*

- Designed the side-scroller level maps, laying out the main combat encounters and boss-fight areas.
- Contributed to combat mechanics, player objectives, encounter flow, progression, and boss encounter design.

Geek'O'Wars – Cyber-Survival Game **Top 3, MLH FrostHacks**
Unreal Engine 4.21 | *C++* | *Shaders*

- Developed third-person combat, aiming and recoil behavior, weapon cooldowns, enemy-wave spawning, and visual effects.

ByteOasis: Code to Escape – Puzzle Adventure **2nd Place, HackRush 2.0**
Unreal Engine 4 | *C++* | *Blueprints*

- Built a programming-inspired adventure involving system restoration, terminal commands, logic puzzles, and progression.

Leadership

President, Elysium Gaming Club – IARE **Sep 2024 – Apr 2025**

- Helped organize Overkill'25, including esports tournaments, developer talks, and student community showcases.

Volunteer and Game Development Mentor, CodeDay **Dec 2024**

- Supported beginners with game design, C++, and Unreal Engine fundamentals; reviewed projects and provided feedback.

Education

Institute of Aeronautical Engineering (IARE) Hyderabad, India
B.Tech in Computer Science and Engineering Expected May 2027
CGPA: 7.5 / 10.0

Certifications

- Unreal Engine 5: The Complete Beginner's Course – Udemy, May 2025.
- CS50x: Introduction to Computer Science – HarvardX, May 2025.
- Python for Data Science – Infosys Springboard, May 2025.
- CS50's Introduction to Game Development – Harvard University, Oct 2024.

Honors and Achievements

- **IGDC 2025:** *City of Aethel* showcase selection.
- **1st Place:** CodeDay 2.0 – *The Interlude*.
- **2nd Place:** HackRush 2.0 – *ByteOasis: Code to Escape*.
- **Top 3:** MLH FrostHacks – *Geek'O'Wars*.

Competitive Gaming

3× Valorant Tournament Winner

- **Consortium 2026** – Winner, Team Backshots.
- **Overkill'25** – Winner, Team Toe Sniffers.
- **Celestium Esports Club Tournament** – Winner, Team Toe Sniffers.

Languages and Interests

Languages: Telugu (native); English (professional); Hindi (working); German (basic, learning).
Interests: Game development, football, photography, guitar, reading, manga, anime, music.